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Description - This document contains highlights of the vendor documentation.

1- HAYES MODEM

If the Modem lights are not lit...

If the modem lights are not lit, plug the modem into another outlet to check power. If this problem occurs with a previously working modem, your modem may have been damaged by a power surge over the electrical or telephone line.

If the RD and/or SD lights come on or are flashing...

Your modem cable may be damaged. Turn the modem off, disconnect the modem cable, and turn the modem cable back on. If the RD or SD light is still on replace modem.

If the CD or HD light remains on ...

If the CD or HD light remains on after you have disconnected, replace Modem.

If the modem will not connect over the line to another modem...

Check the modem and the telephone connections, then try dialing the number on your phone. If you hear a high pitched whistle, then another modem is answering the call, the modems may be having trouble negotiating.

If the modem will not answer an incoming call...

Auto answer may not be enabled Call the online center.

If the modem disconnects unexpectedly...

Try dialing the number again. Also check for loose connections between the modem and computer or between the modem and telephone. If the connections seem secure, you may have had a bad connection. Try again.

2- BIT 3 BUS TO BUS ADAPTER

Chapter 2: Interface Functions

2-1 Executive Overview

Bit 3 Fiber-Optic Interfaces are available in two- and four- cards and modules. Each card or module is one half of a two Interface high performance fiber-optic link between two Bit 3 Adapter cards. Fiber-Optic cable lengths to 2 kilometers (km) are supported.

Model 400-60 are external Fiber-Optic Interface Modules with a chassis, power supply and installed fiber-optic interface card. Bit 3 Fiber-Optic Interface Modules can be used with any Bit 3 Adapter to interconnect two computer systems via fiber-optic cable.

Bit 3's Fiber-Optic Interfaces convert the signals that are normally sent over wire cable to light signals that travels over fiber-optic cables. Because of the inherent capabilities of the fiber-optic cable, the distance between the two computer chassis that are linked by a bit 3 Adapter is greatly extended without impact on the data to be transferred. In addition to extending the distance between chassis, the Fiber-Optic Interfaces provide virtual immunity to the electromagnetic interference.

Fiber-Optic Interface work transparently with Bit 3 Adapter. With Fiber-Optic Interfaces installed, Adapter cards in the chassis at either end of the cable are able to detect if all cables are present and if the adapter card at the opposite end of the link is powered up.

Direct Memory Access (DMA) data transfers are supported by the Fiber-Optic Interfaces.

In most cases the Fiber-Optic Interfaces require no additional software support beyond that required by the Bit 3 Adapter.

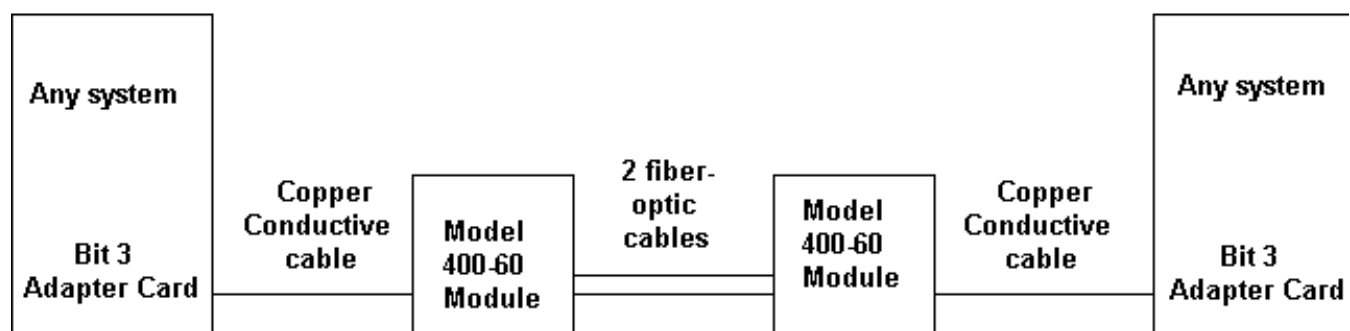
Two Fiber-Optic Interfaces and a Bit 3 Adapter (includes two Adapter cards) are required for a functional link.

Model 400-60 (four Fiber) Modules can link any two Bit 3 Adapter cards.

Model 400-60 Fiber-Optic Interfaces support two duplex fiber-Optic cables for higher data transfer speed and optimal performance. Two duplex fiber-optic cables provide two paths for transmitted data and two for receive data.

The four-fiber models, effectively support Adapters, such as Models 467-1, 477-1, 487-1, 497 and 413-1, that allow high-speed DMA Burst Mode data transfers for applications requiring above 15M Bytes/sec.

2-2 Configuration



BIT 3 ADAPTER WITH FIBER-OPTIC INTERFACE MODULES
ILLUSTRATION 2

2-3 Card Specifications

Transfer Rate:

Model 400-60 30M Bytes/sec.

LED's

Each Fiber Optic Interface Card and Module has two LED status indicators: READY and ACTIVE, READY is lit after the power-up sequence has completed successfully and the Interface is ready to transmit or receive data. ACTIVE is lit when a fiber-optic data transfer is in progress.

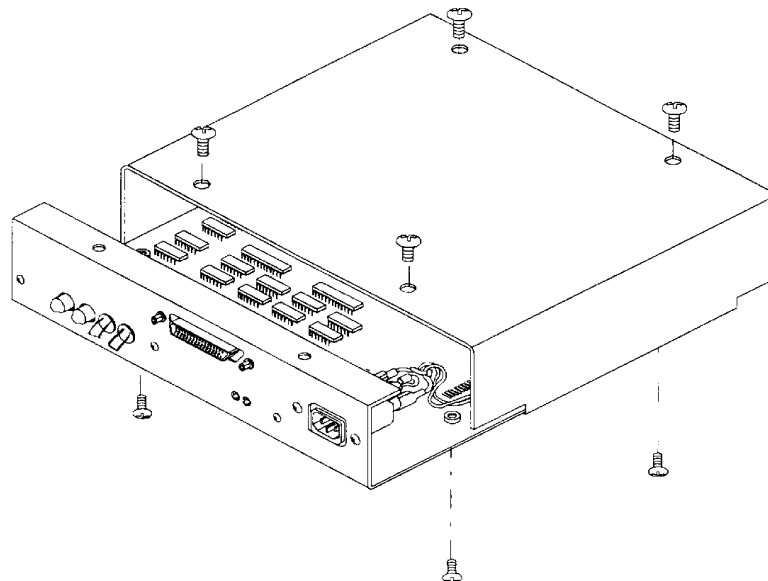
Chassis Dimensions:

Height: 2.5 inches
Width: 12.0 inches
Depth: 10.0 inches

2-4: Hardware Configuration:

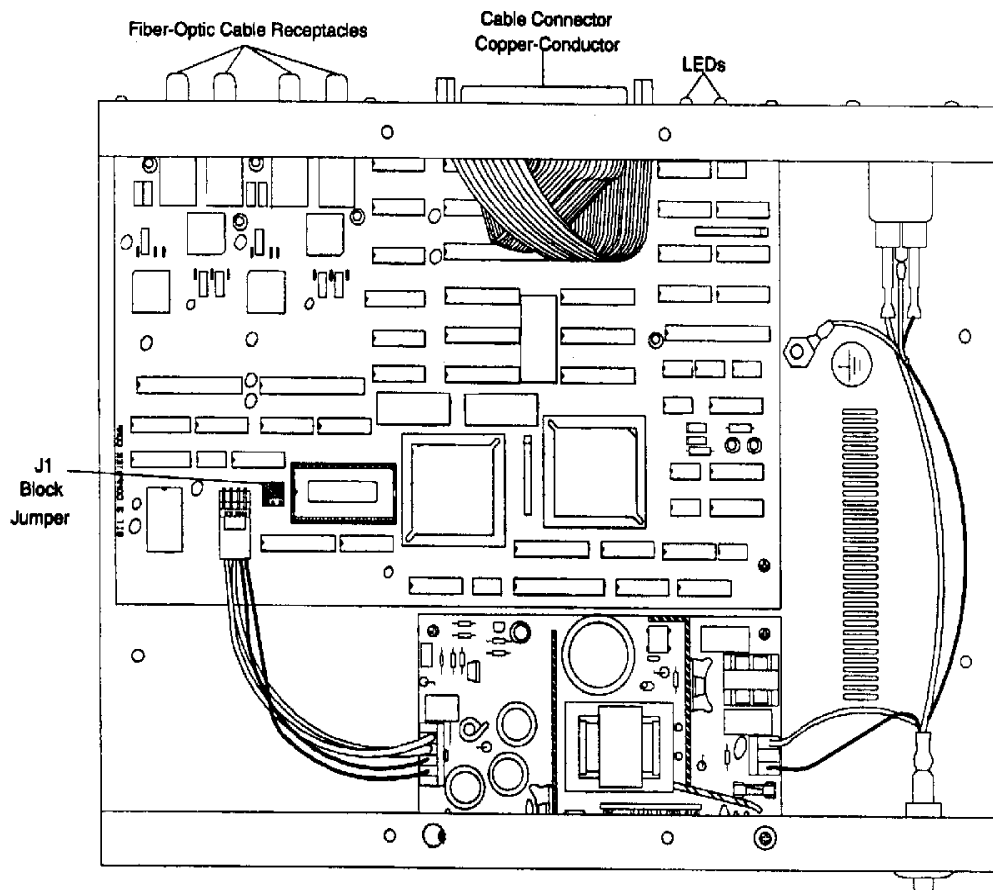
To configure Fiber-Optic Interface Modules

1. Turn power switch to off and detach power cord from Fiber-Optic Interface Module chassis.
2. Remove the screw that secure the chassis cover to the base (four on chassis top and four on the bottom). See Illustration 3.



FIBER-OPTIC INTERFACE MODULE
ILLUSTRATION 3

3. Remove cover and set aside.
4. Locate jumper block J1 on the Interface card see Illustration 4.



FIBER-OPTIC INTERFACE MODULE LOCATION OF JUMPER BLOCK J1, CABLE CONNECTORS & LEDS
ILLUSTRATION 4

5. Jumper block configuration:

- * Jumper 1: must always be in the on position. This jumper is reserved for future use.
- * Jumper 2 : (the Adapter Select Jumper): The position of this jumper is determined by the Bit 3 Adapter with which the Fiber-Optic Interface is used.
- * Jumper 3: Selects the error checking mode.
- * Setting jumper 3 to the on position enables, Enhanced Error Checking.

When configured for the enhanced Error Checking, two error checking circuits are operational. One circuit relies on 4B/5B and 5B/6B data encoding to detect errors. Each data word is encoded before transmission. Upon reception, the code is checked against a list of valid codes. All invalid codes are flagged as errors. Although use of this error checking circuit is fast and provides invalid code detection, it cannot detect errors that cause a valid code to transform into other valid codes.

Additional enhanced error checking, is provided by a second error checking circuit. This second circuit uses a cyclical redundancy check (CRC) algorithm that can detect errors that corrupt a valid code into other codes. The CRC checker provides enhanced error detection: however the additional overhead required reduces performance.

* Setting jumper 3 to the OFF position enables BASIC ERROR CHECKING.

Basic error checking bypasses the CRC checker and uses the code checker (the first error checking circuit); consequently, higher performance is achieved.

6. Replace the chassis cover and screws.

3- PERFORMANCE NOTES

3-1 Introduction

This chapter describes the effects of the Fiber-Optic Interface on the Bit 3 Adapter cards to which they are connected. In all cases, the performance figures below apply only to the Fiber-Optic Interfaces.

3-2 Fiber-Optic Interfaces

Fiber-Optic Interfaces are functionally transparent to the operation of the Bit 3 Adapter. The Interface, however affects Adapter throughput. Both transfer rate and latency are affected.

Transfer rate refers to the sustained DMA throughput the Fiber-Optic Interface is capable of supporting. The larger the block of data transmitted, the closer throughput will be to the transfer rate. However, since the DMA data received by the Fiber-Optic Interface are always accompanied by the header information, actual data throughput in an application is always slightly less than the maximum hardware transfer capability.

Latency refers to the additional time required to complete any transfer, either random access or DMA, over and above the time required by the adapter with no Fiber-Optic Interface. For example, a random access read take an average of 1.6 microseconds longer (Basic Error Checking) with the Fiber-Optic Interface Installed.

The performance statistics below include two sets of figures for each Fiber-Optic Interface Module: Basic Error Checking and Enhanced Error Checking.

3-3 Four Fiber Performance

Basic Error Checking

* Transfer rate up to 30M Byte/sec.

* Latency:

(minimum)	1.5 microseconds
(Average)	1.6 microseconds
(maximum)	1.7 microseconds

Enhanced Error Checking

* Transfer rate up to 30M Byte/sec.

* Latency:

(minimum)	1.8 microseconds
(Average)	1.9 microseconds
(maximum)	2.0 microseconds

3-4 Cable Length Effects

It takes approximately 5 nanoseconds for a signal to travel down one meter of Bit 3 Fiber-optic cable. As a result, approximately 10 nanoseconds of round trip latency is added for each meter of fiber-optic cable used. For example it takes approximately 1 microsecond longer to complete a random access read using a 105 meter cable than it would using a five meter cable.

Appendix A: Glossary

Bit: a single digit in a binary number (0 or 1).

Byte: 8 bits

CRC: Cyclical Redundancy Check. An error checking technique used to ensure the accuracy of transmitting digital code over a communications channel. Transmitted messages are divided into predetermined lengths that are divided by a fixed divisor. The remainder of the calculation is appended onto and sent with the message. At the receiving end the computer recalculates the remainder. If it doesn't match the transmitted remainder, an error is detected.

DMA: Direct Memory Access Transfers. The adapter may be programmed to transfer large blocks of data across the cable to or from the remote chassis, rather than the requiring a processor or move data.

km: Kilometer ; 1000 meters; 3,280.8 feet.

M Byte: Megabyte. Two to the twentieth power (exactly 1048,576) bytes.

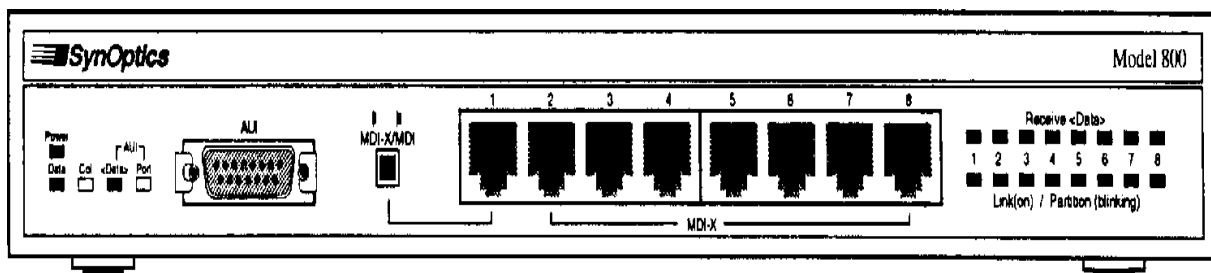
3- ETHERNET HUB MODEL 800

Chapter 3 : Reference

This section describes the Model 800 hub LED indicators and functions, and provides you with information about the hub ports, their connectors, and pin assignments.

3-1 LED Indicators

The Model 800 hub has two sets of LED indicators on the front panel. The LED's on the left side of the hub indicate the overall status of the hub (such as whether or not it is receiving power) and the status of the AUI port. The LED's on the right side of the hub provide status information about the 10BASE-T ports. During the power-up sequence, all LEDs flash for approximately two seconds while the hub performs a lamp test. See Illustration 5.



HUB FRONT PANEL
ILLUSTRATION 5

3-2 AUI Port

The following are the AUI connector, and pin assignments. See Table 2.

TABLE 2
AUI PORT

Pin Number	Signal name
1	Ground
2	CI-A
3	DO-A
4	Ground
5	DI-A
6	+ 12-VDC Return
7	(Not connected)
8	(Not connected)
9	CI-B
10	DO-B
11	Ground
12	DI-B
13	+12 VDC
14	Ground
15	Ground

3-3 10BASE-T Ports

The following table shows the Model 800 RJ-45 port, its connector, and pin assignments. Each of these eight shielded RJ-45 jacks is 10BASE-T port using the standard media-dependent interface with internal crossover (MDI-X) pin assignment shown in the following table. Each port detects the polarity of the circuit connected to RD pins and adjust polarity as necessary to correctly complete this circuit. See Table 3.

TABLE 3
10BASE-T PORTS

Pin no.	MDI-X Signal (function)	MDI Signal (function)
1	RD + (Receive from UTP wire)	TD + (Transmit to UTP wire)
2	RD - (Receive from UTP wire)	TD - (Transmit to UTP wire)
3	TD + (Transmit to UTP wire)	RD + (Receive from UTP wire)
4	Not used by 10BASE-T	Not used by 10BASE-T
5	Not used by 10BASE-T	Not used by 10BASE-T
6	TD - (Transmit to UTP wire)	RD + (Receive from UTP wire)
7	Not used by 10BASE-T	Not used by 10BASE-T
8	Not used by 10BASE-T	Not used by 10BASE-T

3-4 MDI/MDI-X Port 1

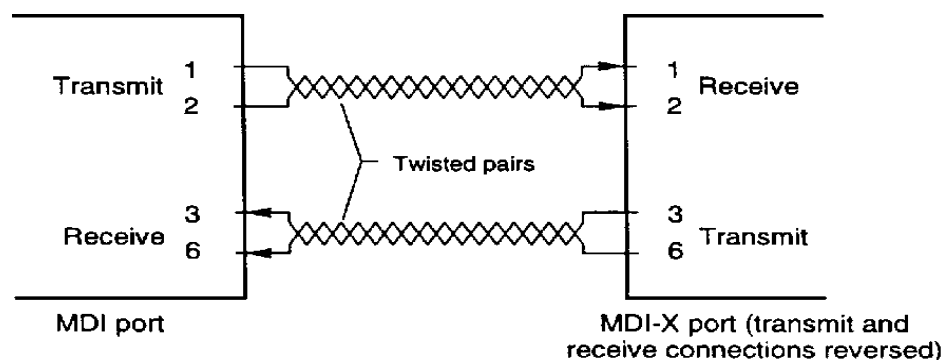
Model 800 hubs can switch 10BASE-T port 1 transmit and receive pins between MDI and MDI-x configurations. An MDI port transmits on RJ -45 pins 1 and 2 and receives on RJ-45 pins 3 and 6.

Ports 2 through 8 are fixed as MDI-X ports, but port 1 can be set as MDI or MDI-X using the MDI-x/MDI switch on the front panel:

*Press in the MDI-X/MDI switch to set port 1 to MDI-X configuration.

*Pop out the MDI-X/MDI switch to set port 1 to MDI configuration.

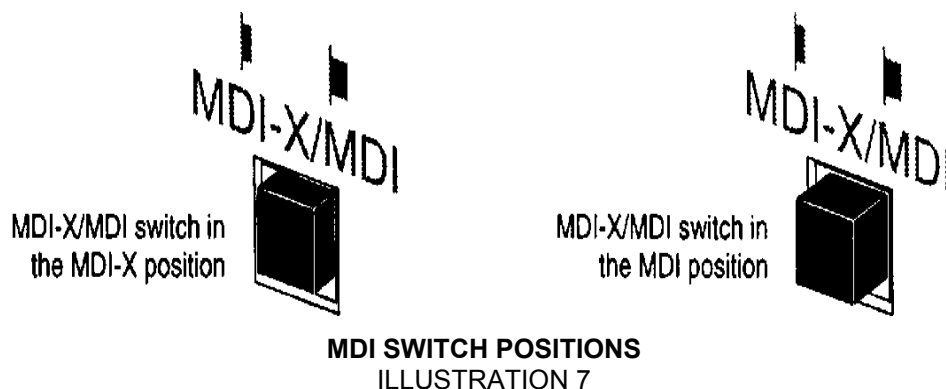
Illustration 6 shows MDI /MDI-X switch in both positions.



MDI AND MDI-X TRANSMIT AND RECEIVE PIN INTERNAL CROSSOVER
ILLUSTRATION 6

This feature allows you to conveniently connect the hub to another hub directly through their RJ-45 connectors. You can use a standard straight through UTP cable between port 1 (set to MDI) and an MDI-X 10BASE-T port on another hub. A crossover cable (nonstandard) is required to connect the hubs directly (MDI-X port to MDI-X port) through UTP cable. Two rules govern all usual RJ-45 connections from port to port on another device.

*A straight through cable (MDI to MDI-X as shown in Illustration 7 is customarily used for connection to a workstation adapter card or transceiver.



*A crossover cable or adapter can be used for directly cabling an MDI-X port on a hub to an MDI-X port on another device See Illustration 8.

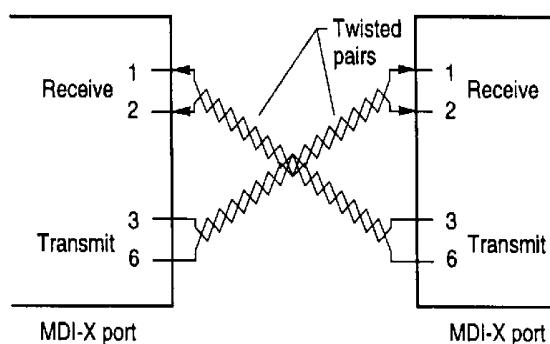


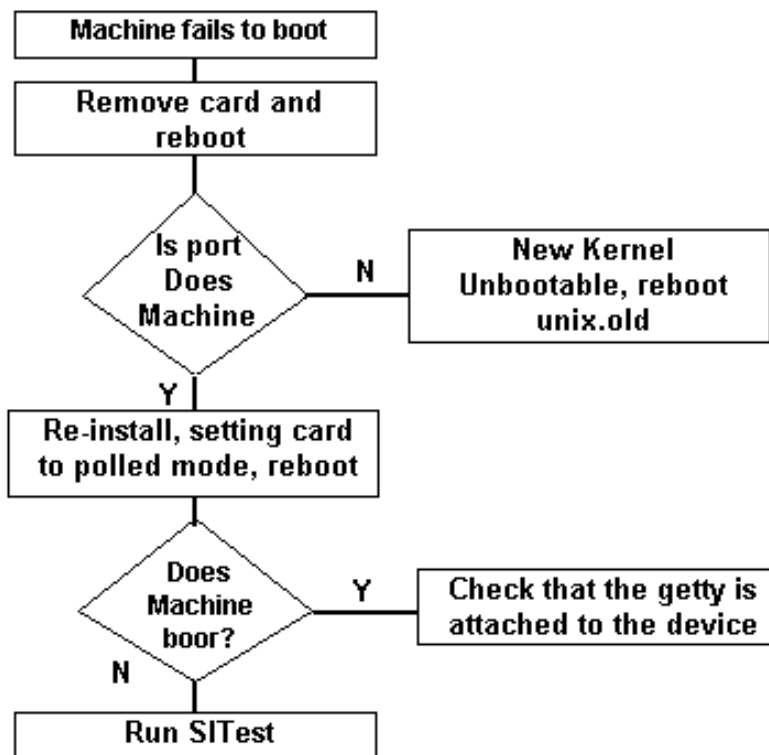
Figure 24. MDI-X to MDI-X external crossover

MDI
ILLUSTRATION 8

4- SPECIALIX

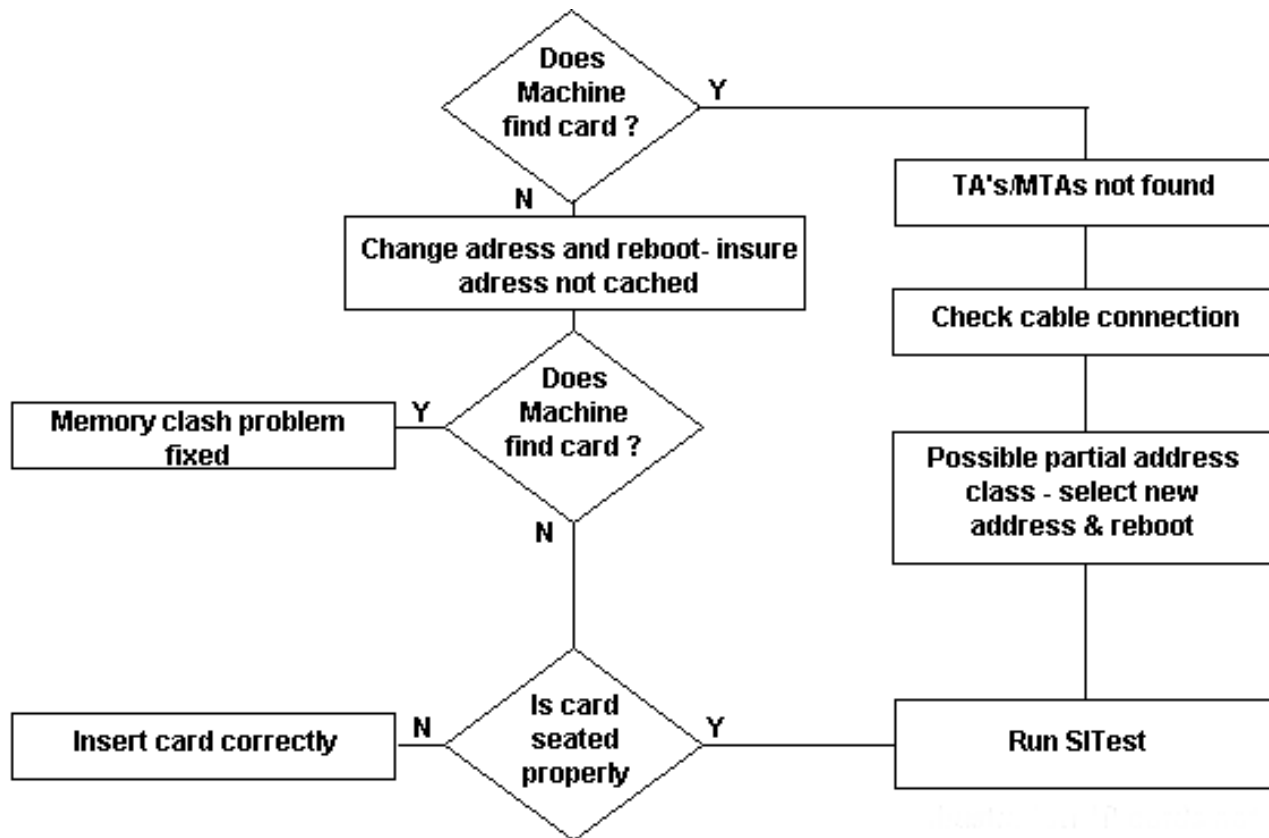
4-1 Troubleshooting Flowcharts

Machine fails to boot, See Illustration 9.



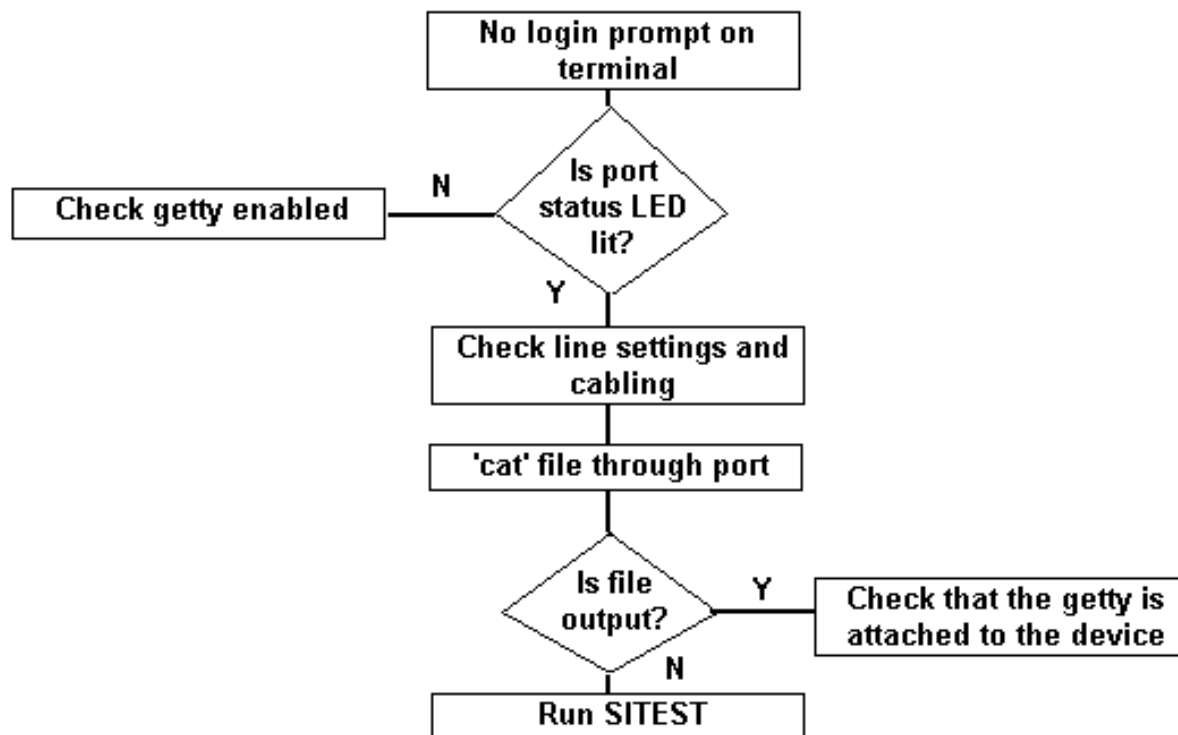
MACHINE FAILS TO BOOT
ILLUSTRATION 9

Cards and or Terminal Adapters not found. See Illustration 10.



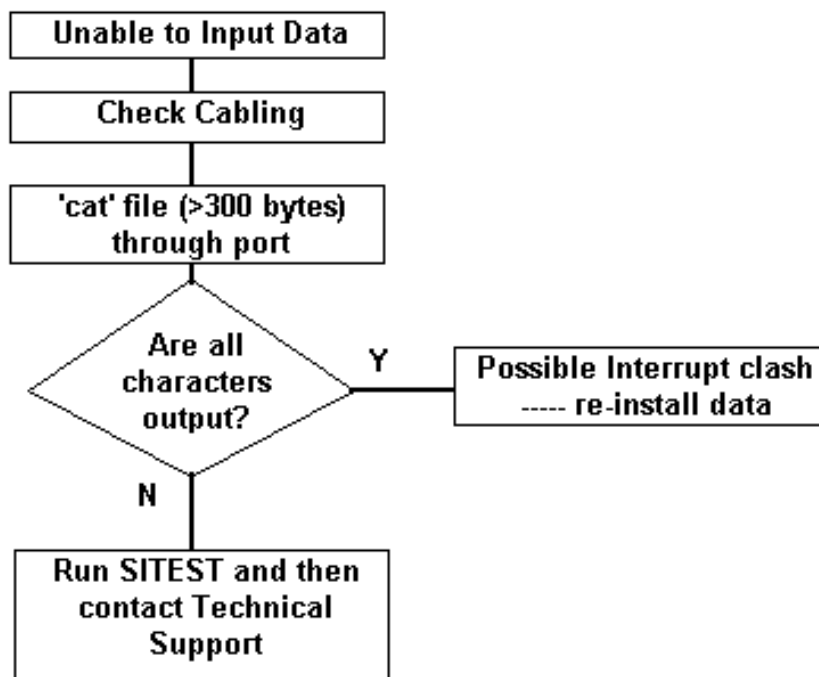
CARDS AND/OR TERMINAL ADAPTER NOT FOUND
ILLUSTRATION 10

Can't obtain login prompt, See Illustration 11.



CAN'T OBTAIN LOGIN PROMPT ON TERMINAL
ILLUSTRATION 11

Unable to input data/login message corrupted, See Illustration 12.

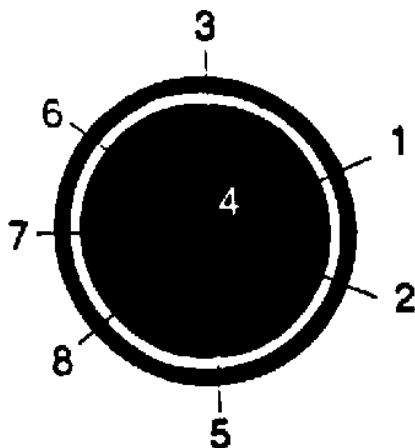


UNABLE TO INPUT DATA / LOGIN MESSAGE CORRUPTED
ILLUSTRATION 12

5- INDIGO WORKSTATION

Cable Pin Out Assignment:

Serial port See Table 4 and illustration 13.

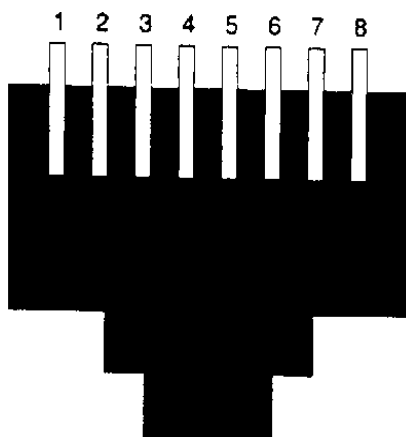


SERIAL PORT
ILLUSTRATION 13

TABLE 4
4D COMPATIBLE PIN ASSIGNMENTS

Pin	Assignment	Description
1	DTR	Data Terminal Ready
2	CTS	Clear to Send
3	TD	Transmit Data
4	SG	Signal to Ground
5	RD	Receive Data
6	RTS	Request to Send
7	DCD	Data Carrier Detect
8	SG	Signal Ground

Ethernet 10-BASE-T See Table 5 and Illustration 14.



PIN LOCATIONS
ILLUSTRATION 14

TABLE 5
ETHERNET 10 BASE T

Pin	Assignment
1	Transmit +
2	Transmit -
3	Receive +
4	(Reserve)
5	(Reserve)
6	Receive -
7	(Reserved)
8	(Reserved)

Ethernet AIU See Table 6 and illustration 15.

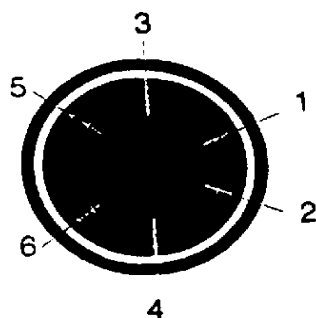


PIN LOCATIONS
ILLUSTRATION 15

TABLE 6
ETHERNET AUI

Pin	Assignment
1	Ground
2	Collision +
3	Transmit +
4	Ground
5	Receive +
6	Ground
7	(Reserve)
8	Ground
9	Collision -
10	Transmit -
11	Ground
12	Receive -
13	+12V
14	Ground
15	(Reserved)

Keyboard and Mouse See Table 7 and Illustration 16.



KEYBOARD AND MOUSE
ILLUSTRATION 16

TABLE 7
KEYBOARD AND MOUSE PINOUT

Pin	Assignment
Keyboard	
1	Keyboard Data
2	(Reserved)
3	Ground
4	Keyboard Power (+5V)
5	Keyboard Clock
6	(Reserved)
Mouse	
1	Mouse Data
2	(reserved)
3	Ground
4	Mouse Power
5	Mouse Clock
6	(Reserved)

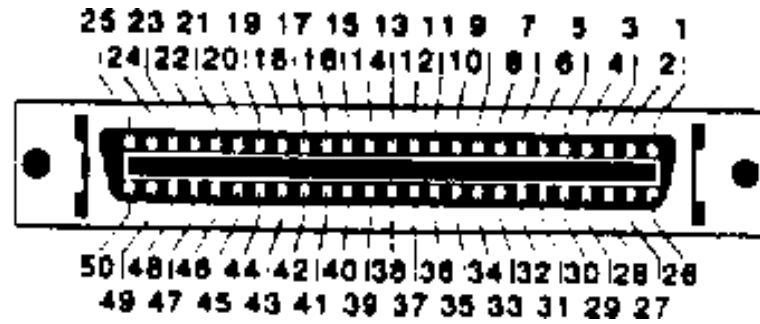
Audio Port See Table 8.

TABLE 8
AUDIO PORTS

Connector	Tip	Ring	Sleeve
Digital Input/Output	Out	In	GROUND
Headphone Output	L	R	GROUND
Line Output	L	R	GROUND
Line Input	L	R	GROUND
Microphone Input	L	R	GROUND

Audio Ports

SCSI Port See Table 9 and illustration 17.



SCSI PINOUTS
ILLUSTRATION 17

TABLE 9
SCSI

PIN	ASSIGNMENT	PIN	ASSIGNMENT
1	GROUND	26	DB(0)
2	GROUND	27	DB(1)
3	GROUND	28	DB(2)
4	GROUND	29	DB(3)
5	GROUND	30	DB(4)
6	GROUND	31	DB(5)
7	GROUND	32	DB(6)
8	GROUND	33	DB(7)
9	GROUND	34	DB(P)
10	GROUND	35	GROUND
11	GROUND	36	GROUND
12	RESERVED	37	RESERVED
13	OPEN	38	TERM/PWR
14	RESERVED	39	RESERVED
15	GROUND	40	GROUND
16	GROUND	41	ATN
17	GROUND	42	GROUND
18	GROUND	43	BSY
19	GROUND	44	ACK
20	GROUND	45	RST
21	GROUND	46	MSG
22	GROUND	47	SEL
23	GROUND	48	C/D
24	GROUND	49	REQ
25	GROUND	50	I/O

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 14, 1998	J. Saperstein	Initial conversion from Toolbook to Word.